

Prologue

HOW TO BE A TEACHER (Part One)

Open the door and walk in. Remain standing. Or maybe you should sit down?

This crowded rectangular room is yours. Right now it has twenty-six chairs with attached desks, a chalkboard, and early-afternoon sunlight pouring through windows onto the tabletops. In a moment, the room will also have twenty-six fifth-graders whose names are printed on the attendance ledger: Richard, Catherine, Anthony, Eddie, Varouna, Giyoo, Awad, Donna Ruth, Tyrone, Ellie, Enoyat, Leticia, Charlotte, Karim, Shanota, Messima, Sandra, Dorota, Ivan, Connie, Illeana, Yasu, Reba, Jumanah, Candice, and Shahroukh.

Your job, according to the state where you happen to live and the school district that pays your salary, is to make sure that, sixty minutes from now, the students have grasped the concept of “rate.” Specifically, if a car is going 55 miles per hour, how far will it have traveled after 15 minutes? How about after 2 hours? By the end of the year, your students should also have mastered fractions, negative numbers, linear functions, long division, ratio and proportion, and exponents. You’re also supposed to teach

them to become good citizens, subtly knitting into your lesson (yes, this math lesson) the principles of democracy. In whatever time is left, remember to help the children vault over any hurdles life has thrown them—racial, economic, parental, intellectual. You must bend reality closer to the dream of the American meritocracy.

Ready?

The door bursts open. With the residual energy of recess, they surge through the coat room, rearranging their clothes and jostling for sips from the water fountain. Here comes Varouna. She is from Kenya, lithe and dark skinned. Giyoo is from Japan. He is 4 feet tall and barely speaks. Catherine is studious and has her hair in braids. Eddie, freckle faced and hyperactive, takes his seat in the back. Tyrone just moved from South Carolina and prefers not to pay attention. He sits closer to you, in the front.

Don't just stand there. Teach something!

Richard sits near the front, next to Tyrone. They're both new to the school this year. On the first day, Richard introduced himself and volunteered that math was his "worse subject."

Half an hour later, the students are all askew, murmuring and chatting with each other. They've been working on a math problem you wrote on the chalkboard while they were out at recess.

Condition: A car is going 55 mph. Make a diagram to show where it will be

- A. after an hour
- B. after 2 hours
- C. after half an hour
- D. after 15 minutes

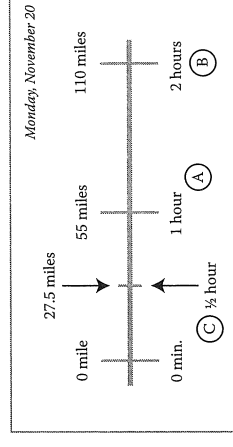
Consider how to get everyone to quiet down. Next to you, on a table, is a small bell. Do you ring it? Perhaps you should raise one

hand and put the other hand over your mouth. Or what about that old line? *When my hand goes up, your mouths go shut.* You go for the bell. Thankfully, it works, and you launch a discussion.

Soon, fifteen minutes have passed, and class is almost over. So far, the students have worked on the problem in small groups of four to six. You have circulated around, peering over shoulders at their varying degrees of success, deciding when to talk and when to nod and when to hold in a laugh, letting it shake inside your chest when a student does something hilarious and adorable. And all of you, together, have reasoned your way through A, B, and C.

On the chalkboard, you've drawn a straight horizontal line, with distance represented on top and time underneath. On the far right is a crosshatch for 110 miles and 2 hours (B); halfway in the middle there is another for 55 miles and 1 hour (A); then there's one more, smaller, crosshatch halfway between 0 and 55: 27.5 miles and $\frac{1}{2}$ hour (C).

It looks like this:



Point to the board. Ask: Can anyone show where the solution to part D should go on the diagram?

Hands shoot up. Then, right in front of you, Richard adds his. You know enough about the others to have an idea of how they understand "rate," or at least an idea of what they will be able to do with the problem. Richard, though, is something of a mystery. After the "worse subject" speech, you collected his math note-

book at the end of each week along with the other students'. But he wrote very little in it and only rarely raised his hand. Now he's volunteering to answer the most difficult part of the question—and you have no idea what he'll say.

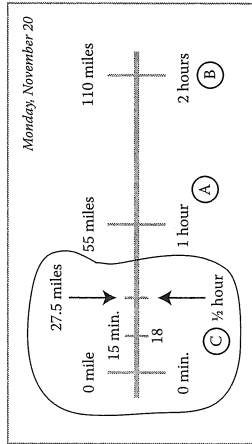
What do you do?

Look at the clock; only 10 minutes left. Do you have time to risk a wrong answer? What about Richard? What if he isn't even close? If he's wrong, will he, an African American boy in a racially diverse classroom, shut down and hesitate to participate again? On the other hand, what message does it send to the others *not* to call on him?

"Richard," you say. He stands up, turning his notebook so he can see it from the board, and walks slowly to the front. Everyone waits, silent.

D: Show where the car going 55 mph will be after 15 minutes.

Reaching for the miles section, on top, he rests the chalk halfway between 0 and 27.5. "15 minutes," he writes. Below, between 0 minutes and $\frac{1}{2}$ hour, he writes, "18." The board looks like this:



"Ummm," he says. "Eighteen."

Huh? Not only has he put time (15 minutes) where distance should go, but he has also proposed another number, 18, that makes no sense. A car going 55 miles an hour could not travel 18 miles in 15 minutes. And what reasonable computation would get you to 18? Not dividing 27.5 by 2, or 110 by 4, certainly, and not anything else related to the numbers on the board either.

What do you do?

You could quickly correct his time-distance reversal, not drawing too much attention to the mistake, on the assumption that it was a careless error. But what if it wasn't? You decide to assume nothing. "Eighteen miles," you venture, "or eighteen minutes?"

Clarify: "You wrote 18 next to minutes. Did you mean 18 miles and 15 minutes?" Richard nods, erases, and rewrites. Now the numbers are flipped: 18 miles, 15 minutes. But there's still that mystifying 18.

What do you do? Should you say, simply and directly, *That's wrong*? What does Richard mean, anyway?

Look at the class. Ask: Can anybody explain what Richard was thinking?

Another jolt of hands. Try to memorize who is asking to speak, and who is making a fan out of his pencils. Remember, you aren't just teaching Richard; the other twenty-five need to be educated too. What are they thinking? Are they learning?

Check the time. Just a few minutes left, but this could take much longer. Maybe better to give up; there's always tomorrow. But look at Richard, who still believes 18 makes sense, who doesn't know what he doesn't know.

Call on studious Catherine. "Ummmmmm," she says. "I disagree with that." She pauses. Then, "Ummm . . ."

Think. She wants to give the correct answer, yet you said, can anybody explain what *Richard* was thinking, not can anybody talk about her own idea. Catherine seems to know she's out of order. That "ummm . . ."—she's eyeing you, looking for permission to disobey.

Do you grant it? Maybe you should. Nod, and the right answer will come—clear and concise, knowing Catherine, and just in time for the end of class. But look at Richard. If quick Catherine, a white girl, jumps in with the save, what effect will that have on

him? On the other hand, if you don't let Catherine continue, how will that affect the rest of the class? In either case, what will the class learn about race, gender, and—oh yeah, math?

On Monday, November 20, 1989, Magdalene Lampert, a stoic, watchful woman with straight blonde hair clipped to the back of her head and more than a decade of teaching experience, made snap decision. She pointed to the 18. "Does anyone *agree* with this answer?" she asked.

The common view of great teachers is that they are born that way. Like Michelle Pfeiffer's ex-marine in *Dangerous Minds*, Edward James Olmos's Jaime Escalante in *Stand and Deliver*, and Robin Williams's "carpe diem"—intoning whistler in *Dead Poets Society*, legendary teachers transform thugs into scholars, dilutes into geniuses, and slackers into bards through brute charisma. Teaching is their calling—not a matter of craft and training, but alchemical inspiration.

Bad teachers, conversely, are portrayed as deliberately sadistic—as with the Sue Sylvester character on *Glee*, congenitally boring (Ben Stein's nasal droner in *Ferris Bueller's Day Off*), or ludicrously dim-witted (Mr. Garrison from *South Park*). These are the tropes of a common narrative, a story I've come to call the Myth of the Natural-Born Teacher."

Even in the rare cases where fictional teachers appear to improve—as happens in *Goodbye, Mr. Chips*, the novel-turned-film, in which a bland schoolteacher named Mr. Chips comes to "sparkle"—the change is an ugly duckling-style unmasking of hidden pizzazz rather than the acquisition of new skill. Others think Mr. Chips has become a "new man," but in fact, we are told, he has only peeled back a "creeping dry rot of pedagogy" to reveal one "sense of humor" that "he had always had."

The idea of the natural-born teacher is embedded in thou-

sands of studies conducted over dozens of years. Again and again, researchers have sought to explain great teaching through personality and character traits. The most effective teachers, researchers have guessed, must be more extroverted, agreeable, conscientious, open to new experiences, empathetic, socially adjusted, emotionally sensitive, persevering, humorous, or all of the above. For decades, though, these studies have proved inconclusive. Great teachers can be extroverts or introverts, humorous or serious, flexible or rigid.

Even those charged with training teachers—the ones who, by definition, should believe teaching can be taught—believe the natural-born-teacher narrative. "I think that there is an innate drive or innate ability for teaching," the dean of the College of Education at Chicago State University, Sylvia Gist, told me when I met with her in 2009. The consensus seems to be, you either have it or you don't.

Before I met Magdalene Lampert, I ascribed to this view as well. My teacher friends seemed born for the blackboard. I could see it in their personalities and in how much they cared—one's earnest, unabashed sensitivity; another's confident, playful devotion. Gregarious, charming, and theatrical, they commanded attention wherever they went. No wonder they decided to teach, while I—shamefully serious, allergic to goofiness, prone to skepticism—became a journalist. They had the magical quality of "teacherness"—what Jane Hannaway, the director of the National Center for Analysis of Longitudinal Data in Education Research and a former teacher, described to me as "voodoo."

When I first met Magdalene, her talent was obvious, and it did, at first, look like voodoo. It was the winter of 2009, twenty years after she taught fifth-grade math to Catherine and Richard, and she was now a professor at the University of Michigan's School of Education. We sat in her sun-soaked office, at the far

end of a long table, looking at the work of a fifth-grader named Brandon.

In the course of solving a problem about the price of party ribbons, Brandon had mistakenly declared that $\frac{7}{12} = 1.5$. What, Magdalene asked me, could have made him think that?

This was probably the first time Magdalene read my mind, which is what she does after asking a question. She lowers her eyelids slightly, purses her lips, and peers into your soul. I had no idea how Brandon could have come up with 1.5, and she knew it. But instead of giving me the answer, she wanted me to think about what might make sense (just as, back in 1989, she had wanted Richard to think about his answer, 18). She drew a long-division sign, that “house” that I remembered from fifth grade.

She placed the numbers in the wrong spots: 12 under the house and 7 outside of it, to the left, as if we were asking how many times 7 went into 12 rather than how many times 12 went into 7. Putting 7 into 12, a student would find that it went in once, with a remainder of 5 ($12 - 7$). “1 R 5,” he would have written, in the language of fifth grade.

When we looked at Brandon’s paper, that is exactly what we saw: a house over 12, with 7 on the outside, and then “1 r 5” written next to it in green marker. Brandon, Magdalene explained, must have mistakenly translated his “1 r 5” into 1.5. (The answer is actually 1 and $\frac{5}{12}$.)

It seemed like a magic trick—how quickly Magdalene moved from noticing a problem to diagnosing its source. Instead of just looking at the final wrong answer, she had translated Brandon’s notes—almost nonsensical to me—into a logical (if flawed) path, skipped backward through his thinking, and located the original point of misfire. It took her no more than a minute.

And what about all the other errors Brandon could have made as he struggled to find the price of those ribbons? What about the

mistakes scattered through his classmates’ papers, not to mention all the ones that weren’t there, but could have been? This, after all, was just the work of one class, taken from one day out of the year, in one grade and one subject, by one student. I watched, captivated, as Magdalene worked through more papers, reading backward through the minds of the children, each prone to his own unique mistakes.

But the more I learned about Magdalene and her teaching, the more I saw that what looked like mind reading was in fact the result of extraordinary skill, not inborn talent. Her success did not depend on her personality, which—inward, pensive, and measured—was in many ways the opposite of Hollywood’s mythic teachers. Instead, Magdalene’s success relied on a body of knowledge and skill that she had spent years acquiring. Teaching, as she practiced it, was a complex craft.

Magdalene showed me that the illusion of the natural-born teacher is at best a polite version of the old adage attributed to George Bernard Shaw: “He who can, does. He who cannot, teaches.” By imagining teaching as a “voodoo” mixture of personal charisma and passion, we are saying, essentially, *He who has intelligence, does. He who has charm, teaches*. I have come to think that this is a dangerous notion. By misunderstanding how teaching works, we misunderstand what it will take to make it better—ensuring that, far too often, teaching doesn’t work at all.

“Aha!” Magdalene Lampert’s decision not to correct Richard had paid off—partly, anyway.

His nonsensical answer, that the car traveling at a speed of 55 miles per hour would go 18 miles in 15 minutes, remained on the board. But after Magdalene asked the class whether anyone agreed with his answer, enduring an uncomfortable pause when nobody said anything, Richard finally broke the silence.

"Can I change my mind?" he asked her. Instead of 18, he wanted to "put thirteen and a half or thirteen point five."

Better! The calculation he should have made is that, since 55 miles corresponds to 60 minutes, and half of 55 miles, 27.5, corresponds to 30 minutes, then a quarter of 60 minutes would be half again: 13.75. He was close.

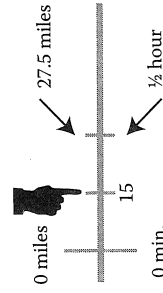
But Magdalene still didn't understand why he'd first said 18. She needed to know exactly what had gone wrong inside his head. Pointing to the place on the chalkboard where Richard had originally written "18," she asked him why he'd changed his mind.

He was back in his seat. "Because," he said, "eighteen plus eighteen isn't twenty-seven."

"Aha!" she said, permitting herself a minor celebration.

He had it—at least, most of it. Keeping her hand on the board so that it covered up the old wrong answer, 18, Magdalene pivoted so that her body faced Richard and the rest of the class. She wanted everyone to hear what she had to say. Richard had gone from stumbling to coming up with the beginnings of a proof—a mathematical argument for why 18 couldn't be the answer—and she wanted to draw everyone else's attention to his work.

This is what the board showed:



Whatever goes in this spot, she said, has to be a number that, when doubled, comes close to 27.

In the back of the room, students started murmuring. "Not close!" one shouted out. Another student threw up his hand.

Magdalene took note but did not make a move just yet. She thought about 27. The board, of course, said the correct number

of miles that is half of 55—27.5, not 27. If he had been shooting for precision, Richard would have tried to find a number that, when doubled, equaled 27.5, not 27. But if they were talking about a real car, making a real trip, would it matter if he calculated a distance of 13.5 miles, rather than 13.75? It might, and it also might not. Still, learning to make approximations was an important skill, and Magdalene was happy with Richard's performance. He was estimating—proving, even—thinking mathematically.

She did not want to make Richard think he'd made a mistake, but she also wanted to help him and the rest of the class reach the exact answer. After all, if she hadn't wanted the students to deal with the tricky matter of having to divide 27.5 into two pieces, she could have picked a rounder speed, like 60 miles an hour. Then the math would have been nice and clean. But one of her objectives for the year was to have students learn to convert between decimals and fractions, and to divide each of them in their heads. She had picked 55 because she wanted the class to struggle with exactly this problem, in exactly this way.

How to acknowledge Richard's good work but also, at the same time, correct it? She surveyed the growing field of raised hands. Anthony, a small boy whom Magdalene knew loved to talk, was waving his hand in the air. Awad, a quiet boy with neat, cursive handwriting, had his hand up too. Who would keep up her ambiguous tone: accepting Richard's answer, but expanding on it? She chose Awad.

Paradoxically, the institution most susceptible to the fallacy of the natural-born teacher is our country's public school system. And that's despite the fact that alarm—always high—over the disappointing level of our national teaching quality has recently reached a fever pitch.

"From the moment our children step into a classroom," Barack

Obama said in 2007, “the single most important factor determining their achievement is not the color of their skin or where they come from; it’s not who their parents are or how much money they have. It’s who their teacher is.” Obama was then a presidential candidate; in office, his position only strengthened. Today, thanks to policies that his administration has advanced, school districts across the country are undergoing ambitious efforts to reinvigorate their teaching force. The debate about these reforms is fierce; many people, including many teachers, oppose Obama’s efforts. But their objection is not usually with his premise. They agree that teachers matter and that the quality of their work should be improved. What they dispute is how to enact the change.

One argument—Obama’s—prescribes improvement by way of *accountability*. The problem with American education, this line of thinking goes, is that we have for too long treated all teachers the same: they get the same pay raises, the same evaluations, and the same job protections whether they inspire their students like Robin Williams or stultify them like Ben Stein. But the fact is that some teachers are good and some are bad. Some help children learn while others set them back.

“They have 300,000 teachers in California,” Obama explained in a speech in 2009. “The top 10 percent are 30,000 of the best that are out there. The bottom 10 percent are 30,000 of the worst out there. The problem is, we have no way to tell which is which.” This, he went on, “is where data comes in.” By measuring which teachers are successful and which aren’t, we can reward the phenomena and discard the duds, thereby improving the overall quality of the teaching force. Following Obama’s prescription, revamped teacher evaluation systems are now being rolled out across the country, along with rewards and punishments that will affect teachers’ careers.

The other argument—call it the *autonomy* thesis—prescribes

exactly the opposite. Where accountability proponents call for extensive student testing and frequent on-the-job evaluations, autonomy supporters say that teachers are professionals and should be treated accordingly. Like lawyers or doctors, they will improve only if they are given the trust, respect, and freedom they need to do their jobs well. Lately, proponents of this argument have been drawing comparisons to Finland. There, a recent report by the Chicago Teachers Union described, “teaching is a respected, top career choice; teachers have autonomy in their classrooms, work collectively to develop the school curriculum, and participate in shared governance of the school.” In Finland, the report concludes, teachers “are not rated; they are trusted.”

As descriptions, both arguments—accountability and autonomy—contain a measure of truth. Teachers do lack some of the freedom they need to teach well, and they also lack adequate feedback. But as prescriptions, actual suggestions for how to improve teaching, the arguments fail. Neither change, on its own, will produce better teachers. Basic math makes the problem with accountability clear: Discard the bottom 10 percent and, as Obama said, that’s thirty thousand teachers who will need to be replaced. And that’s just in California. Nationally, the number is more than ten times that. Autonomy, meanwhile, is an experiment that many schools have tried for years, and still seen teachers struggle.

Neither accountability nor autonomy is enough, in other words, because both arguments subscribe to the myth of the natural-born teacher. In both cases, the assumption is that good teachers know what to do to help their students learn. These good teachers should either be allowed to do their jobs or be held accountable for not doing them, and they will perform better. Both arguments, finally, rest on a feeble bet: that the average teacher will figure out how to become an expert teacher—alone.

This bet is especially audacious, considering the large number of people involved. More people teach in this country than work at McDonald's, Wal-Mart, and the U.S. Post Office combined. In New York City, where I live, a corps of teachers seventy-five thousand strong makes up a workforce roughly the same size as Apple's global employee base. As Amy McIntosh, the former chief talent officer of New York City's Department of Education, pointed out, in all the five boroughs there is no building where all seventy-five thousand teachers could gather at a single time. Not even Yankee Stadium (capacity 50,287).

Of the fields to which teaching is commonly compared—those that require a college degree and are considered of reasonably high social value—none come close to matching the number of employees that teaching has. Consider a bar graph displaying the number of Americans in different professions. The shortest bar represents architects: 180,000. Farther over, slightly higher, come psychologists (185,000) and then lawyers (952,000), followed by engineers (1.3 million) and waiters (1.8 million). At the top stand the big three: janitors, maids, and household cleaners (3.3 million); secretaries (3.6 million); and, finally, teachers (3.7 million). An ongoing swell of baby boomer retirements is expected to force school systems to hire more than three million new teachers between 2014 and 2020. As the departing teachers wave goodbye to their students, they will take all their experience and skill out the door with them. These new hires will have to replace them.

One December night in 2009, I watched as hundreds of the people hoping to become teachers packed an auditorium at Chicago's Cultural Center, home of the world's largest stained-glass Tiffany dome, to hear from the city school system's director of recruitment. There were no seats available, and the sea of humanity was as diverse as it was vast. There was a cross-eyed woman with white hair and a disheveled look. There was a dreadlocked

recent college graduate with hair dangling below his belt. There were many dozens of young midwestern ladies with their mothers, taking careful notes. There was a small woman in a Christmas sweater with ornaments sewn into quadrants, including a Velcro nameplate stuck on her left breast: RACHEL.

But even if everyone in the auditorium had signed up to teach—the mothers along with their daughters—the crowd still would not have filled all the available teaching slots. Each year, the city of Chicago hires two thousand new teachers. That year, the economic downturn had lowered the number below its average. But the district still needed six hundred new teachers. Nationwide, nearly four hundred thousand new teachers start work at public and private schools every year.

When all these people take their place in front of classrooms across the country—from the overcrowded trailers in Queens, New York, to the humid, ranch-style spaces serving Alabama Native American reservations, to the breezy, open-air classrooms of Cerritos, California—what will they do? What *should* they do? And how can we make sure all of them do the best possible job?

The cold truth is that accountability and autonomy, the two dominant philosophies for teacher improvement, have left us with no real plan. Autonomy lets teachers succeed or fail on their own terms, with little guidance. Accountability tells them only whether they have succeeded, not what to do to improve. Instead of helping, both prescriptions preserve a long-standing culture of abandonment. Steven Farr of Teach For America described this culture by telling me about the first time his assigned mentor came to observe his class. The mentor was just doing her job, but when she walked in, she apologized, as if for some voyeuristic intrusion. Teaching, she told him, is “the second-most private act.” She’d rather not be caught watching someone else do it.

The sociologist Dan Lortie, in his classic work *Schoolteacher*,

describes the teaching profession in the language of Victorian-era sex: a private "ordeal." Lortie traces the fundamental loneliness to the days of the one-room schoolhouse, when teachers worked in isolation because the other adults (and some of the children) were busy farming. These days, there are more personnel and more students associated with each classroom, but each teacher still faces a room full of pupils alone.

What do teachers do? They do what any of us would do. They make it up.

That day in November, Magdalene Lampert's gamble to call on Awad—carefully calculated, in her case—paid off. Awad played exactly the role she had hoped, correcting Richard's imprecision about 13.5 without trampling over his accomplishment in getting there.

"Ummm," Awad had said with his typical deliberation. "I think it's thirteen point seventy-five." Richard kept his composure, and in the minutes that followed, Magdalene untangled a series of teaching problems. She called on Anthony, who had been waving his hand in the air, but didn't let him go on too long and even distilled a clear, concise idea from his confusing, if enthusiastic, speech.

She then gave the floor to a girl, Ellie, balancing the gender of speakers and thereby minimizing the idea that only boys can do math, which paved the way for an astonishing performance by another girl, Yasu, who constructed a sophisticated logical proof that recalled Richard's original insight about the relationship between doubling and halving. All this had happened in just a few minutes. But now, it was beyond time for class to end. The teacher who was to take over the room after math ended stood at the back of the classroom, giving Magdalene a look.

"You know what I think?" she said to the class, nodding at the

teacher in the back. "I think that we are going to schedule a little time on remainders and division. 'Cause I think we are getting a little mix—We are mixing up a lot of ideas here and we don't have time to go into them."

She paused again. She wanted to give anyone who might be deeply confused one last chance to ask a question. The students sat before her, their math notebooks still open in front of them: Richard in the front, Awad in the back, Catherine to her right. All of them would be there tomorrow too, and the next day and the next and the next, until summer.

"Okay?" Magdalene asked, turning the statement into a slight question—a door just on its way to being closed. No one said anything.

Okay.

Both sides of the "teacher quality" debate tend to depict the challenge as a transfer problem—how to help unsuccessful, often low-income students (like the ones I cover as a reporter in New York City) to access the experiences enjoyed by their more affluent peers (like the ones I had attending public school in the manicured Washington, DC, suburb of Montgomery County, Maryland).

The accountability argument holds that suburban schools have the best teachers because, with rich coffers and newer, prettier buildings, they are able to lure top talent. To rebalance this unequal distribution, Obama has supported measures to tempt high-quality teachers back to school districts serving poorer populations. Proponents of the autonomy argument, meanwhile, contend that teachers working with the poor have paradoxically received the *least* freedom and the most restrictive working environments. Make their schools look more like those enjoyed by the children of the wealthy, and they will be able to prosper.

Again, neither description is wrong, but as prescriptions, both are incomplete. Teachers at affluent public schools do enjoy, on average, better working conditions and more flexibility. But they are also victims of the natural-born-teacher hypothesis. Indeed, the more I learned about successful teaching, the more I realized how rare it is, even in the schools with the most resources.

Not long ago, exploring the closet of my childhood bedroom in Maryland, I discovered a pink, cardboard filing cabinet that held my elementary school papers. In the best classrooms I visited as a reporter, children were reading and writing by kindergarten. My pink filing cabinet did not have a kindergarten file. What would I have put in it? That year, I did not know how to read.

The first-grade papers, meanwhile, bore little resemblance to the careful work I saw in classrooms run by excellent reading teachers. The file contained words copied from worksheets and not much original writing. By January, I had reached my peak level for the year: I was able to fill a collection of construction paper sheets, stapled together and labeled—in an adult's handwriting—"Writing Journal," with lifeless one-sentence entries:

Water is fun.

I havt cold weather.

I like toeys.

I like Sarah.

I like sissors.

It was fun at the show.

I haet work. It is to eyse.

No wonder I didn't read until first grade. The work *was* too easy, and as a result I didn't learn.

Even later, in the special "gifted" programs I attended from fourth grade on, how much had my teachers really taught me?

Some changed my life forever, helping me fall in love with journalism, calculus, and even quantum physics. But what about the others? Besides that fluke physics year, my memories from science classes were mainly of lethargic fruit flies. And only in my last year of high school did I figure out that history had to do with evidence and arguments, as well as memorizing state capitols and the dates of irrelevant wars.

I attended some of the fanciest public schools in the country (my school district, in Montgomery County, Maryland, has an average household income in the country's top ten), yet the teaching I received was just as inconsistent as at the schools I later visited in Newark, New Jersey, the Bronx, and San Francisco.

Yet, while I have come to see that the scope of America's education challenge is much larger than I ever imagined, I have also begun to see a path through which the challenge might be tackled. For every case I have found of the natural-born-teacher fallacy hampering progress—and I have found a lot, stretching far into the past—I have found another case of a person who thought differently.

Take Colonel Francis Parker. Born in 1837 in New Hampshire, the son and grandson of teachers, Parker believed that teaching well required intense study. Teaching was, he said, "the greatest art in all the world"; learning to do it well could take a lifetime. But it didn't take Parker long to learn that this was an unpopular view. After serving honorably in the Civil War, he was offered several prestigious jobs that would take him away from the classroom. "When I said that I was going to be a school teacher [instead]"—during the war, he had spent nights before the campfire, planning future lessons in his head, and he did not intend to give them up—"my friends were very much disgusted with me." Even another teacher called him a fool.

His fellow teachers, Parker was finding, mirrored the general

public. Many of them didn't think about their work as a craft they needed to study. Later, when Parker took over a struggling school for teachers in Chicago, most of the city (including some teachers) wondered whether the school should exist at all. "The fact of the matter is, the conviction that young men and women should be trained for their work in order to teach little children existed only here and there," one of his colleagues said. "The general public was against it."

Parker died before seeing his dream of resuscitating the school and its reputation fully realized. He was, said the rabbi Emil G. Hirsch in his eulogy, "another Moses," destined to behold his promised land only from afar. The same fate befell Parker's successor at the University of Chicago, the philosopher John Dewey. Expanding on Parker's vision, Dewey had written eloquently about the "science of education" he hoped to develop—how it would help prevent the immeasurable "waste" that comes from letting great teachers' secrets live and die with them. "The only way by which we can prevent such waste in the future," he wrote, "is by methods which enable us to make an *analysis* of what the gifted teacher does intuitively, so that something accruing from his work can be communicated to others." The science wasn't to be—at least, not yet. For half a century after he made it, Dewey's prescription lay in hibernation, the victim of the same forces that Parker had pressed against.

But though Parker and Dewey both died before seeing "educational Palestine," as Emil Hirsch called it, their vision did not. Today, the natural-born-teacher illusion lives on, but thanks to Magdalene Lampert and a growing group of educators like her, so does Parker and Dewey's dream.

The educators include some people like Magdalene, longtime teachers who later became a unique breed of researcher, studying their own craft while they worked to pass it on to others.

They include, as well, people who echoed Magdalene's conclusions without ever meeting her—sometimes deliberately (as happened in the 1980s, on an island six thousand miles away from her fifth-grade classroom), other times not (as in the case of the movement of entrepreneurial educators that emerged a decade later). Together, these educators still constitute a minority. But for a variety of reasons, their chances of building Francis Parker's educational Palestine are better than any other time in history.

This book is their story. It is also the story of teaching, that hilarious and heartbreaking theater that unfolds between children and teachers every day. The work that, when done well, with trained skill, can induce in a student a near-magical feeling: the trembling sensation of beholding a new idea where nothing existed before. It begins with one of the first pioneers—a shy, industrious man named Nathaniel Gage.